

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF APPLIED SCIENCES AND HUMANITIES
SESSIONAL TEST-1

Set-1

Course: B. Tech.
Session: 2024-25
Subject: Engineering Chemistry
Max. Marks: 20

Semester: I
Section: A, B, C, D
Subject code: BAS-102
Time: 60 Minutes

Q. No. :	1	2	3	4	5	6
CO. No. :	4	4	4	4	4	4
Bloom's Level	K2	K2	K2	K2	K2	K2

Note: This Question paper contains six questions and all questions are compulsory.

Section A (2 Marks)	
Q1. Write the composition of Bio-Gas.	2 Mark
Section B (6 Marks)	
Q2. The water sample has been found to contain the following impurities expressed in ppm: $\text{Ca}(\text{HCO}_3)_2 = 40.5$, $\text{Mg}(\text{HCO}_3)_2 = 36.5$, $\text{CaSO}_4 = 34.0$, $\text{CaCl}_2 = 27.75$, $\text{MgSO}_4 = 30$, $\text{NaCl} = 10.0$ Calculate the temporary, permanent and total hardness of water sample.	3 Mark
Q3. Explain the Reverse Osmosis process of desalination of brackish water.	3 Mark
Section C (12 Marks)	
Q4. Explain the zeolite process for the removal of hardness of water.	4 Mark
Q5. A sample of coal was analysed as follows: Exactly 2.000 g was weighed into a silica crucible. After heating for one hour at 110°C , the residue weighed 1.975 g. The crucible next was covered with a vented lid and strongly heated for exactly seven minutes at $950 \pm 20^\circ\text{C}$. The residue weighed 1.328 g. The crucible was then heated without the cover, until a constant weight was obtained. The last residue was found to weigh 0.205 g. Calculate the percentage results of the above analysis.	4 Mark
Q6. With the help of a neat diagram, explain the working of Bomb calorimeter.	4 Mark

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40+14x2

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF APPLIED SCIENCES AND HUMANITIES
SESSIONAL TEST-2

Course: B. Tech.
 Session: 2024-25
 Subject: Engineering Chemistry
 Max. Marks: 20

Semester: I
 Section: A, B, C, D
 Subject code: BAS-102
 Time: 60 Minutes

Q. No. :	1	2	3	4	5	6
CO. No. :	4	4	5	5	5	5
Bloom's Level	K2	K2	K2	K2	K2	K2

Note: *This Question paper contains six questions and all questions are compulsory.*

Section A (2 Marks)		
Q1.	Why is it conventional of express the hardness of water in terms of CaCO_3 at the international level?	2 Mark
Section B (6 Marks)		
Q2.	A 0.90 g sample of a solid fuel was completely combusted in the oxygen using bomb calorimeter. The rise in temperature of water in calorimeter was 3.8°C . Calculate the high calorific value of the fuel. If water taken in calorimeter is 4000 g and water equivalent of calorimeter is 385 g. Also calculate the low calorific value. Given % of H in fuel =1 and latent heat of steam = 587cal/g	3 Mark
Q3.	Write a short note on Polymer Blends.	3 Mark
Section C (12 Marks)		
Q4.	What are conducting polymers? Write the structure of polypyrrole and polythiophene. Discuss about conducting nature of these polymers.	4 Mark
Q5.	Explain the preparation, properties and applications of following polymers: (i) Bakelite (ii) Nylon 6 (iii) Dacron	4 Mark
Q6.	Discuss any five applications of Grignard reagents.	4 Mark

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF APPLIED SCIENCES AND HUMANITIES
SESSIONAL TEST-3

Course: B. Tech.
 Session: 2024-25
 Subject: Engineering Chemistry
 Max. Marks: 20

Semester: I
 Section: A, B, C, D
 Subject code: BAS-102
 Time: 60 Minutes

Q. No. :	1	2	3	4	5	6
CO. No. :	2	3	2	1	1	3
Bloom's Level	K2	K2	K2	K2	K2	K2

Note: This Question paper contains six questions and all questions are compulsory.

Section A (2 Marks)		
Q1.	Calculate the number of NMR signals to the following compounds: (i) $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—OH}$ (ii) $\text{CH}_3\text{—CO—CH}_3$ (iii) $\text{C}_6\text{H}_5\text{—C}_2\text{H}_5$ (iv) CH_3CHO	2 Mark
Section B (6 Marks)		
Q2.	Calculate the cell potential of the following cell: $\text{Zn}_{(s)} / \text{Zn}^{2+} (0.1\text{M}) \parallel \text{Cu}^{2+} (1.75\text{M}) / \text{Cu}_{(s)}$ at 25°C Given: $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34\text{V}$	3 Mark
Q3.	Write a short note on optical isomerism in compounds without having asymmetric carbon.	3 Mark
Section C (12 Marks)		
Q4.	Draw the molecular orbital diagram of NO and N_2 molecules. Calculate their bond order and predicts their magnetic behavior.	4 Mark
Q5.	What are liquid crystals? Explain various types of thermotropic liquid crystals.	4 Mark
Q6.	Describe the various steps involve in the manufacturing of Portland cement by rotary kiln method.	4 Mark

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF APPLIED SCIENCES AND HUMANITIES
MAKE UP SESSIONAL TEST-I

Course: B. Tech.
Session: 2024-25
Subject: Engineering Chemistry
Max. Marks: 20

Semester: I
Section: A, B, C, D
Subject code: BAS-102
Time: 60 Minutes

Q. No. :	1	2	3	4	5	6
CO. No. :	4	4	4	4	4	4
Bloom's Level	K2	K2	K2	K2	K2	K2

Note: This Question paper contains six questions and all questions are compulsory.

Section A (2 Marks)		
Q1.	Write the composition of Bio-Gas.	Mark
Section B (6 Marks)		
Q2.	The water sample has been found to contain the following impurities expressed in ppm: $\text{Ca}(\text{HCO}_3)_2 = 16.2$, $\text{Mg}(\text{HCO}_3)_2 = 14.6$, $\text{CaSO}_4 = 13.6$, $\text{CaCl}_2 = 16$, $\text{MgSO}_4 = 12$, $\text{NaCl} = 20$ Calculate the temporary, permanent and total hardness of water sample.	3 Mark
Q3.	Explain the Reverse Osmosis process of desalination of brackish water.	3 Mark
Section C (12 Marks)		
Q4.	Explain the zeolite process for the removal of hardness of water.	4 Mark
Q5.	A sample of coal contains: $\text{C} = 75\%$, $\text{O} = 30\%$, $\text{H} = 6\%$, $\text{S} = 0.5\%$, $\text{N} = 0.49\%$, $\text{Ash} = 0.9\%$ Calculate GCV and NCV of the Coal.	4 Mark
Q6.	With the help of a neat diagram, explain the working of Bomb calorimeter.	4 Mark